

NICE Toolbox: Nonverbal Interpersonal Communication Exploration Toolbox

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Software

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Summary

Understanding human communication is a long-standing goal in psychological research. People express themselves through many verbal and nonverbal channels: posture, gesture, gaze, facial expression, and speech. Measurement and annotation of these signals has traditionally required time-consuming manual work by trained human labellers. The **Nonverbal Interpersonal Communication Exploration (NICE) Toolbox** is a free, open-source Python framework that automatically measures these signals by combining existing state-of-the-art computer vision and speech-processing methods into a single software package. It includes support for multi-camera systems, interchangeable detection algorithms, 2D and 3D visualization of detector outputs, evaluation on the user's custom data, and export to third-party software. With the release of this tool, we aim to make data exploration of interpersonal communication signals scalable, reproducible, and accessible for psychological research.

Statement of need

Labelling and annotation of nonverbal behaviour signals from video recordings is one of the key methods of psychological research on interpersonal communication ([Bakeman & Quera, 2011](#)). Traditionally, this is done manually by trained student assistants or professional labellers rewatching videos multiple times. The process is slow, difficult to scale, and results are hard to make consistent across annotators ([Vinciarelli et al., 2009](#)). This also limits study dataset size, due to labelling budget and time constraints.

Computer vision and speech-processing models exist for each of the underlying signals (body pose, gaze, speech transcription, and others), but they are distributed as separate research libraries with different software dependencies, input/output formats, and configuration conventions. Combining them into a working analysis workflow requires machine-learning and software-engineering expertise, which usually lies outside the scope of a psychology research project.

Furthermore, choosing among existing models to determine the most suitable one for a specific study can be difficult for most psychological researchers. Data recording setups vary widely in camera placement, lighting, participant distance, appearance, interaction style, and data formats. Algorithm rankings on public benchmarks often do not transfer to a specific study, so researchers must compare candidate models and hyperparameter configurations on their own data before committing to one for their study ([D'Amour et al., 2022](#)). As of yet, no available

41 algorithm framework offers such an integrated performance evaluation pipeline while allowing
42 the combination of different models to capture multimodal data streams.

43 NICE Toolbox is a Python framework for psychology researchers that unifies data processing
44 and model inference across a large set of machine-learning algorithms. It uses a flexible
45 configuration system for defining custom user experiments, visualizations, evaluations, and
46 integration with third-party software. The measured signals are exported as tabular data for
47 further statistical analysis as part of a psychological study.

48 State of the field

49 Manual annotation tools such as ANVIL ([Kipp, 2001](#)) and ELAN ([Wittenburg et al., 2006](#))
50 have long been the standard for behavioural coding in psychological and linguistic research.
51 They provide advanced user interfaces for frame-accurate labelling and are well integrated
52 with the workflows of the communities that use them, but the annotation itself remains a
53 time-consuming and labour-intensive process.

54 Automated open-source signal measurement tools usually cover only one specific channel.
55 Motion Energy Analysis (MEA) ([Ramseyer, 2020](#)) quantifies frame-to-frame pixel change
56 as a proxy for body movement and has been widely used to study interpersonal synchrony
57 in psychotherapy. OpenFace ([Baltrusaitis et al., 2018](#)) and Py-Feat ([Cheong et al., 2023](#))
58 are open-source libraries for facial analysis used in psychological research. CrisperWhisper
59 ([Wagner et al., 2024](#)) and WhisperX ([Bain et al., 2023](#)) are widely used for audio transcription.
60 Combining these tools into a multimodal analysis pipeline is done by the researcher, who must
61 adapt output formats, coordinate systems, and sampling rates using custom post-processing
62 scripts.

63 One effort to centralise these tools in a single hub is EnvisionBox ([EnvisionBox, 2026](#)). It
64 is a community-driven platform that curates open code modules, tutorials, and datasets for
65 multimodal analysis in social-interaction research. Rather than providing a single unified
66 software, it functions as a shared storage for code examples demonstrating model inference
67 and data analysis.

68 Commercial platforms such as FaceReader ([Lewinski et al., 2014](#)) and iMotions ([iMotions
69 A/S, 2025](#)) provide integrated multimodal analysis with polished user interfaces, but their
70 proprietary licensing, closed algorithms, and costs limit reproducibility and accessibility for
71 academic research. Their coverage is also bounded by what the vendor supports: signals or
72 models not offered by the platform cannot be added by the user.

73 The closest comparison to NICE Toolbox is DISCOVER ([Hallmen et al., 2025](#)), a server-based
74 back-end for computationally driven behaviour analysis. Users drive it through its graphical
75 front-end, the NOVA annotation tool ([Heimerl et al., 2019](#)), to explore content, inspect recordings,
76 and refine annotations.

77 Compared to DISCOVER and other alternatives, NICE Toolbox focuses on automated large-scale
78 dataset processing without human intervention or refinement. Each user-created configuration
79 defines a computational pipeline in which detector outputs from different modalities can
80 feed into one another. The same pipeline supports second-level measurements built on top
81 of these outputs (for example, kinematics from body pose, or interpersonal distance from
82 two-person tracking). Rather than committing to a single model, NICE Toolbox provides an
83 interchangeable collection of models with a unified output structure. This allows users to
84 evaluate algorithms and choose the one that works best for their specific task. NICE Toolbox
85 supports multi-camera recordings and fuses per-camera detections into a single 3D space
86 through triangulation, whereas most existing tools either accept only one camera per recording
87 or leave per-camera outputs unfused.

Software design

NICE Toolbox's design is based on three core principles: (1) to provide a configuration-rich experiment design that lets non-programmers run a full multimodal analysis without writing code, (2) to maintain consistent data structures (components) across different detectors, and (3) to reuse established open-source models and libraries rather than reimplementing them (including MMPose (Contributors, 2020), SAM 3D Body (Yang et al., 2026), ETH-XGaze (Zhang et al., 2020), Py-Feat (Cheong et al., 2023), and others).

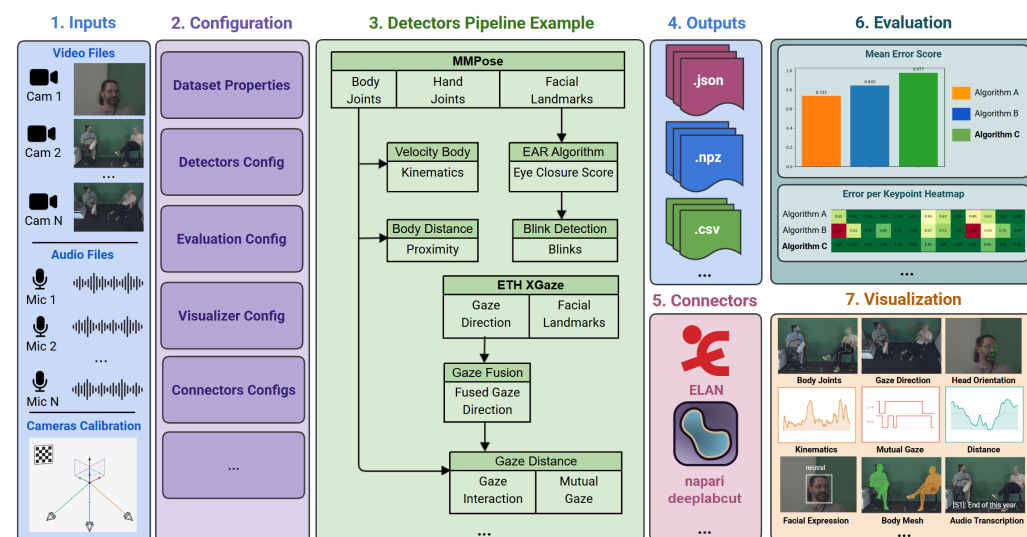


Figure 1: NICE Toolbox experiment example. Multi-camera video with optional audio and calibration files (1) are combined with user-supplied TOML configuration (2) that describes the dataset, requested detectors, evaluation, visualisation, and connectors. A detector pipeline (3) connects primary models (e.g., MMPose for body pose estimation, ETH-XGaze for gaze direction) with second-level detectors (kinematics from body velocity, proximity from inter-body distance, blinks from eye-closure ratios, and mutual gaze from fused gaze directions across cameras). Results are saved per subject in tabular formats (4), exported to external annotation tools such as ELAN (5), and evaluated across candidate algorithms on the user's own recordings (6), with metrics such as mean error score per algorithm and per-keypoint error heatmap. Interactive 2D/3D visualisation (7) shows final results for inspection and diagnosis.

The toolbox is organised into modules (see the example experiment at Figure 1) with separate responsibilities: **detectors** runs the computer vision and audio models on recordings, **evaluation** computes configurable metrics based on the user-specific dataset, **visualizer** provides the interactive rerun (Rerun Development Team, 2024) 3D visualizer, and **connectors** handles import and export to third-party tools such as ELAN and napari-deeplabcut (DeepLabCut, n.d.) for ground-truth annotation.

Users control the software through TOML configuration files that describe algorithm parameters, evaluation metrics, visualizer layouts (see Figure 2), and dataset structure. Configurations are validated against schema definitions before execution, so errors are caught before a long-running pipeline starts. To keep configuration concise and flexible, NICE Toolbox supports variables and named wildcards inside the configuration system. Detector algorithms additionally support template inheritance, so users can define a parameter set once and derive variants that override only the fields that differ; for example, running the same pose estimator with different confidence thresholds, or comparing two backbones that share the same camera setup and I/O components.

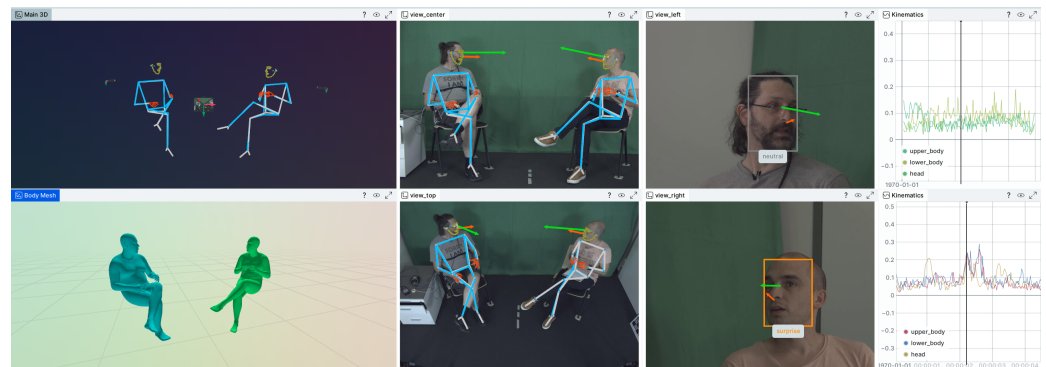


Figure 2: An example layout of the interactive NICE Toolbox rerun visualizer, showing detector results from a dyadic interaction: triangulated body joints, body mesh, gaze direction, head orientation, and synchronised kinematics time series.

110 Detectors can use the output of other detectors as their input, allowing the construction of
 111 a directed computational graph. The order of execution is defined by topological sorting of
 112 nodes. Nodes outputting the same components are interchangeable and configurable by the
 113 user. For example, a user can replace the MMPose model body_joints component with the
 114 SAM-3D-Body body_joints component and use it as input for the kinematics detector; the
 115 kinematics detector stays the same, but its input can be easily replaced.

116 Most of the detectors run in their own isolated virtual or conda environment. The main
 117 reason is that ML libraries often pin conflicting versions of shared dependencies (e.g., PyTorch,
 118 CUDA); isolating each detector lets us use published models as-is without patching them into a
 119 shared dependency tree. The main pipeline launches each detector as a subprocess in its own
 120 environment, passes configuration through a serialized file, and reads the detector's outputs
 121 back from disk in the unified data format.

122 Research impact statement

123 NICE Toolbox has been used to extract behavioural cues for psychological research at the Max
 124 Planck Institute for Intelligent Systems and by other collaborators. In work led by LMU Munich,
 125 the toolbox was used to quantify movement synchrony from pose-estimation-based kinematic
 126 displacement measures across 32 videos of 16 parent-child dyads undergoing psychotherapy,
 127 distinguishing genuine interactive coupling from coincidental covariation and comparing free-
 128 play with structured-play conditions (Buaría et al., 2026). In a separate study on emotional
 129 and behavioural responses to microaggressions in human-AI interaction at work, the toolbox
 130 provided multimodal behavioural measurements (facial expression, gaze, posture) across 482
 131 interaction sequences recorded from six synchronised cameras, collected alongside self-reported
 132 responses (Singh et al., 2026).

133 NICE Toolbox is released under the GNU Affero General Public License v3 (AGPL-3.0) and is
 134 publicly available on GitHub and Docker Hub, with documentation, tutorials, and an example
 135 dataset.

136 AI usage disclosure

137 Generative AI tools (Anthropic Claude Sonnet 4.6 and Opus 4.8) were used during the
 138 development of this software and during manuscript editing. All AI-generated code was
 139 reviewed by the original contributor and by a second human reviewer in the merge-request
 140 process. All AI-assisted text was reviewed and edited by the authors, who take full responsibility
 141 for the final content.

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